

PROJECT AND TEAM INFORMATION

Project Title

RakshaNetra - NHAI Worker Safety and Emergency Response System

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

NHAI construction and infrastructure projects involve a large number of workers, staff, vehicles, and different work locations. Managing their records, verifying identities, maintaining site security, and responding quickly during emergencies can be difficult when information is maintained manually or in separate systems. During accidents or other emergencies, delays in identifying workers and arranging suitable rescue support can increase the risk to human life.

The motivation behind RakshaNetra – NHAI Worker Safety and Emergency Response System is to develop a simple and organized system that can help manage worker information, verify authorized personnel, monitor safety-related activities, and support emergency response. The system can maintain records of workers and staff along with their roles and identification details, making important information easier to access when required.

Another important aspect is emergency management. When an accident occurs, the system can record the emergency, classify it according to severity, and help prioritize emergency cases. It can also assist in identifying available rescue teams and selecting an appropriate response based on the situation.

Through this project, we aim to combine Data Structures in C and Object-Oriented Programming in C++ to create a practical safety management solution. The project will demonstrate how programming concepts can be applied to a real-world problem while keeping the system simple, efficient, and easy to use.

State of the Art / Current solution

Currently, safety at NHAI project sites is managed through safety rules, regular inspections, worker training, safety audits, and emergency procedures. Worker details, attendance, identification records, accident reports, and other safety information are usually maintained by project teams or contractors. During an emergency, the responsible staff identify the incident, check the available information, and arrange medical or rescue support. However, when information is stored in different records or handled manually, finding the required details can take time. This may make emergency response slower and less organized.

Worker records, identity verification, security activities, emergency details, and rescue team information may also be managed separately. This makes it difficult to get all the necessary information quickly during an emergency.

RakshaNetra aims to provide a simple centralized system that brings these functions together. It will maintain worker and staff records, verify authorized personnel, manage emergency information, prioritize serious incidents, and help assign suitable rescue teams. The system will make important information easier to access and help make safety and emergency management more organized and efficient.

Project Goals and Milestones

The main goal of RakshaNetra – NHAI Worker Safety and Emergency Response System is to create a simple system that helps manage workers, staff, security, and emergencies at NHAI project sites.

Project Goals :

- *Store and manage details of workers and staff.*
- *Check the identity of workers and staff before allowing access.*
- *Keep records of site entry and exit.*
- *Record accidents and other emergency situations.*
- *Give higher priority to serious emergencies.*
- *Keep details of available rescue teams.*
- *Help assign a suitable rescue team during an emergency.*
- *Make worker and emergency information easy to find.*
- *Use DSA concepts in C and OOP concepts in C++ to build the system.*

Milestones :

The project will first start with collecting the requirements and deciding the main features of the system. Next, we will plan the different sections of the project and how they will work. Then, we will create the worker and staff management section, followed by identity verification and entry/exit records. After that, the emergency section will be developed to record emergencies and give priority to serious cases. The rescue team section will then be added to store team details and assign teams during emergencies. Once all sections are ready, they will be connected into one complete system. Finally, the system will be tested, errors will be fixed, and the final report, presentation, and project demonstration will be prepared.

Project Approach

The project will be developed as a simple software-based system using C and C++. The main aim is to create an organized system for managing workers, staff, security information, emergencies, and rescue teams at NHAI project sites.

First, we will identify the requirements of the system and divide the project into different modules. The main modules will include Worker and Staff Management, Identity Verification, Security Management, Emergency Management, and Rescue Team Management. Each module will handle a specific part of the system.

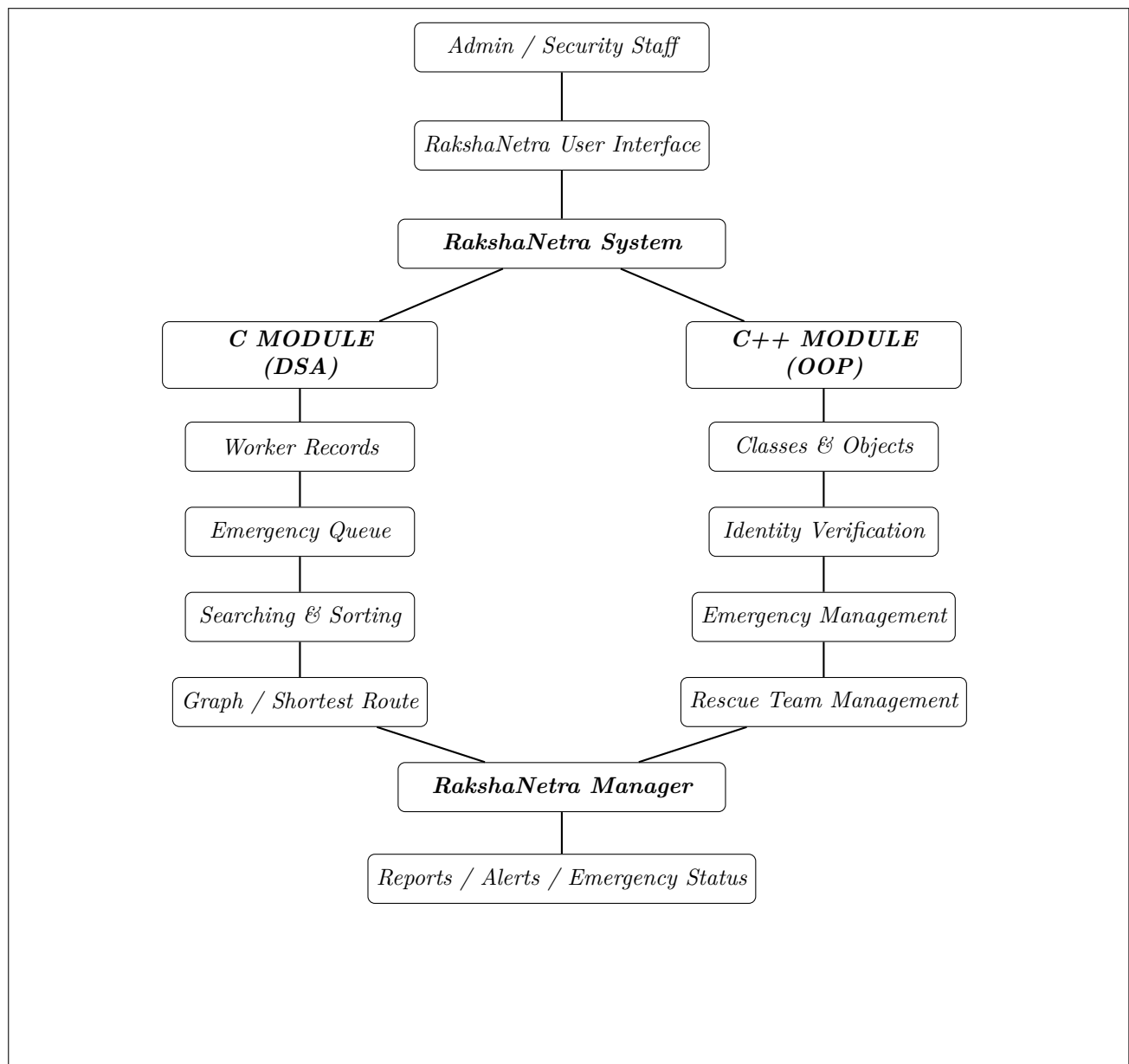
C will mainly be used for implementing Data Structures and Algorithms. Data structures such as arrays, linked lists, queues, priority queues, and graphs will be used wherever required. Searching and sorting techniques will also be used to manage and find records efficiently. A graph and suitable algorithm can be used to represent project locations and help find a suitable route for rescue teams.

C++ will be used to apply Object-Oriented Programming concepts. Classes and objects will be created for workers, staff, emergencies, and rescue teams. Concepts such as encapsulation, inheritance, and polymorphism will be used to make the system organized and easier to manage.

The project will have a simple menu-based interface so that users can easily add, update, search, and view records. Emergency cases can be registered and given priority according to their severity. Available rescue teams can then be managed and assigned to suitable emergencies.

File handling will be used to store important records so that data is not lost when the program is closed. The project will be developed and tested using Visual Studio Code on a computer. Sample data will be used for testing. Finally, all modules will be connected and tested together to create a complete and working system.

System Architecture (High Level Diagram)



Project Outcome / Deliverables

The main outcome of RakshaNetra – NHAI Worker Safety and Emergency Response System will be a simple and organized approach to managing worker safety and emergency information at NHAI. The following deliveries of project are:

- * Worker and staff record system which will keep the identification details, roles and basic information.*
- * Verification system to identify the authorized workers and staff at construction site.*
- * A safety monitoring module to record and manage the safety related activities and incidents.*
- * An emergency management module to record the emergencies and classify based on severity.*
- * A priority system which will help to identify the critical emergencies which will need faster attention.*
- * A rescue team management feature to maintain the available rescue teams information and assist in assigning the suitable team during emergencies.*
- * A simple C and C++ based implementation which will demonstrate the use of Data Structures and Object-Oriented Programming.*

The final system will provide an organized way to access the important safety information and support faster decision-making during emergencies.

Assumptions

It is assumed that the worker and staff details entered in the system are correct and updated when needed. Each worker will have a unique ID for easy identification. Only authorized staff will add or change the records. It is assumed that emergency details such as the location, type, and seriousness of an accident will be entered correctly. Rescue team information will also be kept updated. The system is developed as a simple project model to demonstrate how worker safety and emergency information can be managed. It is not meant to replace real emergency services or existing NHAI systems.

References

- 1. cppreference – C Language Reference — Used for C language syntax, arrays, structures, functions, pointers, memory management, and standard library concepts.*
- 2. cppreference – C++ Language Reference — Used for classes, objects, inheritance, polymorphism, exception handling, and other C++ concepts*
- 3. cppreference – C++ Standard Library — Used for STL containers, algorithms, iterators, and input/output facilities.*
- 4. H. Yang et al., “Design and implementation of an identification system in construction site safety for proactive accident prevention,” Accident Analysis and Prevention, 2012. — Used as a reference for worker identification, authorized personnel, and construction-site safety.*
- 5. S. Rana, S. Sengupta, S. Jana, R. Dan, M. Sultana, and D. Sengupta, “Quick Accident Response System (QARS) Using Internet of Things (IoT),” 2020 Fifth International Conference on Research in Computational Intelligence and Communication Networks (ICRCICN), Bangalore, India, 2020. — Used as a reference for accident reporting, emergency alerts, accident location, and emergency response.*